

MA1513 Exam (Exemplify) Solution

AY2025/2026 Semester I

MCQ/FIB (15 Questions; 3 marks each)

Q1. Given the general solution of a non-homogeneous system $Ax = b$:

$$\begin{cases} x = s - t + 1 \\ y = s \\ z = t \end{cases}.$$

Which of the following are true?

A. $\begin{cases} x = t - s \\ y = t \\ z = s \end{cases}$ is a general solution of $Ax = 0$

B. $\begin{cases} x = s - t + 1 \\ y = s + 1 \\ z = t + 1 \end{cases}$ is a general solution of $Ax = b$

C. $\begin{cases} x = s - t \\ y = s + 1 \\ z = t + 1 \end{cases}$ is a general solution of $Ax = 0$

D. $\begin{cases} x = 1 \\ y = t \\ z = t \end{cases}$ is a general solution of $Ax = b$

E. $\begin{cases} x = 1 - s + t \\ y = s \\ z = t \end{cases}$ is a general solution of $Ax = b$

Explanation

From the given general solution (X) $\begin{cases} x = s - t + 1 \\ y = s \\ z = t \end{cases}$ of $Ax = b$, we can split it into the

“parameter part” (I) $\begin{cases} x = s - t \\ y = s \\ z = t \end{cases}$ and the “constant term part” (II) $\begin{cases} x = 1 \\ y = 0 \\ z = 0 \end{cases}$.

(I) is a general solution of the homogeneous system $Ax = 0$, and (II) is a particular solution of $Ax = b$.

Option A is (I) with the notations of s and t swopped. So it is a general solution of $Ax = 0$.

Option B is adding (I) with (b) $\begin{cases} x = 1 \\ y = 1 \\ z = 1 \end{cases}$, which is another particular solution of $Ax = b$

(substitute $s = 1$ and $t = 1$ in (X).) This gives another general solution of $Ax = b$.

Option C is adding (I) with (c) $\begin{cases} x = 0 \\ y = 1 \\ z = 1 \end{cases}$, which is a particular solution of $Ax = 0$

(substitute $s = 1$ and $t = 1$ in (I).) This gives another general solution of $Ax = 0$.

Option D is a special case of option b by taking $s = t$. Since it has only one parameter, it is not a general solution of $Ax = b$.

Option E is similar to the given general solution (X), but has the signs of s and t swopped in component x . This will not give a solution of $Ax = b$.

Q2. Given the matrix

$$A = \begin{pmatrix} 1 & 1 & * \\ a & b & * \\ b & a & * \end{pmatrix}$$

where a, b are non-zero numbers and $a \neq \pm b$.

Which of the following can be the third column of A if $\det(A) = 0$?

A. $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ a+b \\ a+b \end{pmatrix}$

C. $\begin{pmatrix} 1 \\ a \\ b \end{pmatrix}$

D. $\begin{pmatrix} 0 \\ b \\ a \end{pmatrix}$

E. $\begin{pmatrix} 1 \\ 2a-b \\ 2b-a \end{pmatrix}$

Explanation

Option A: substitute the column into the third column of matrix A and expand (inductively), we will get 0 determinant.

Option B: substitute the column into the third column of matrix A and expand (inductively), we will get $b^2 - a^2$, which is not 0.

Option C: substitute the column into the third column of matrix A will give two identical columns. We will get 0 determinant.

Option D: substitute the column into the third column of matrix A and expand (inductively), we will get $b^2 - a^2$, which is not 0.

Option E: This column is a linear combination of the first two columns of A : $2C_1 - C_2$. Since the columns are linearly dependent, the resulting matrix has determinant 0.

Q3. A is an $n \times n$ matrix such that the homogeneous system $Ax = 0$ has a non-trivial solution. Which of the following statements are true?

- A. Nullity of A is not 0.
- B. The last column of the reduced row echelon form of A must be a non-pivot column.
- C. The n rows of A are linearly dependent
- D. The eigenspace of A corresponding to eigenvalue 0 has dimension 1.
- E. There is an $n \times 1$ matrix b such that the non-homogeneous system $Ax = b$ is inconsistent.

Explanation

Option A: $Ax = 0$ has non-trivial solution means that it has infinitely many solutions. So the nullspace of A contains more than just the zero vector. Hence its nullity cannot be 0.

Option B: A is a singular matrix and its RREF must have non-pivot columns. However, the non-pivot columns may not be the last column.

Option C: Since A is singular, its rows must form a linearly dependent set.

Option D: Since A is singular, 0 is an eigenvalue of A. However, the dimension of the eigenspace E_0 need not be 1.

Option E: Since A is singular, its RREF will have a zero row. This means that, for some non-homogeneous system $Ax = b$, its augmented matrix $[A \mid b]$ may have RREF with zero's on the left but non-zero constant term. This gives an inconsistent system.

Q4. Given the matrix

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}$$

Which of the following sets form a basis for the column space of A?

- A. $\left\{ \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$
- B. $\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \right\}$
- C. $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$
- D. $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$

$$E. \left\{ \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix} \right\}$$

Explanation

The RREF of A has two pivot columns. So the column space of A has dimension 2 (number of basis vector is 2). So option A cannot be a basis for the column space.

For options B to E, each set contains two linearly independent vectors. The only thing left to be checked is whether the two vectors belong to the column space.

Option B: the two vectors are the 2nd and 3rd columns of A. So they belong to the column space.

Option C: the two vectors both have third components equal to 0. This set can only give linear combinations with third component 0, and hence cannot build (generate) those vectors in the columns of A.

Option D: The second vector is the 3rd column of A. The first vector is a linear combination of the 1st and 2nd column of A: $C_2 - C_1$

Option E: Not easy to do by inspection. Check that the columns of A are not linear combinations of the two vectors in E.

Q5. Which of the following sets are linearly independent?

- A. $\{(10, 11, 12), (20, 21, 22), (30, 31, 32)\}$
- B. $\{(1, 3, 2, 4), (4, 2, 3, 1), (1, 1, 1, 1)\}$
- C. $\{(0, 1, 2, 3), (0, 2, 3, 1), (0, 3, 1, 2)\}$
- D. $\{(0, 0, 0, 1), (0, 0, 1, 5), (0, 1, 5, 1), (1, 5, 1, 3)\}$
- E. $\{(2, 3, 5), (7, 11, 13), (17, 19, 23), (29, 31, 37)\}$

Explanation

You can easily check this by stacking the vectors in each option as rows of a matrix, then check whether the RREF has any zero rows. If yes, then the set is linearly dependent; if no, then the set is linearly independent.

Q6. Which of the following linear spans are equal to $\text{span}\{(1, 0, 1), (0, 1, 0)\}$?

- A. $\text{span}\{(1, 1, 1)\}$
- B. $\text{span}\{(1, 1, 0), (0, 0, 1)\}$
- C. $\text{span}\{(1, 0, 1), (0, 1, 0), (1, 1, 1)\}$
- D. $\text{span}\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$
- E. $\text{span}\{(2, 1, 2), (1, 2, 1)\}$

Explanation

First let's compare dimensions. The linear span in the question is of dimension 2, because the two vectors in the span are linearly independent.

Option A has dimension 1 (spanned by a single vector), while option D has dimension 3 (spanned by three independent vectors). So they cannot be equal to our linear span.

Although option C has three vectors, the 3rd vector is the sum of the first two vectors, hence is redundant and can be removed from the span. This will give the same set as our linear span.

Option B: check that the two vectors are not linear combinations of (1,0,1) and (0,1,0).

Option E: check that the two vectors are linear combinations of (1,0,1) and (0,1,0).

Q7. FIB - Let $v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$, $v_4 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ and $w = \begin{pmatrix} 3 \\ 4 \\ 5 \\ 6 \end{pmatrix}$,

The coordinate of w with respect to $S = \{v_1, v_2, v_3, v_4\}$ is given by $[w]_S = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ where a, b, c, d are ...

Ans: $a = 2, b = 1, c = 0, d = 3$

Explanation

This can be easily solved as a linear system $Ax = w$ where $A = [v_1 \ v_2 \ v_3 \ v_4]$.

Q8. FIB - Given $u_1 = (1 \ 1 \ 0 \ 2)$, $u_2 = (2 \ 0 \ 1 \ 1)$, $u_3 = (1 \ 0 \ 2 \ 1)$.

Suppose $v = (a, b, c, -3)$ is orthogonal to u_1, u_2 and u_3 . Then a, b, c are ...

Ans: $a = 1, b = 5, c = 1$

Explanation

Take the dot product of v with u_1, u_2 and u_3 and equate them to 0.

This will give a linear system with 3 equations and 3 variables a, b, c . Solve the system to get the answer.

Q9. FIB - Given $T: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ a linear transformation such that

$$T\left(\begin{pmatrix} 3 \\ 5 \\ 2 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \text{ and } T\left(\begin{pmatrix} 4 \\ 4 \\ 2 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \text{ Then } T\left(\begin{pmatrix} 1 \\ 7 \\ 2 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \text{ where } a, b, c \text{ are ...}$$

Ans: $a = 7, b = 4, c = 1$

Explanation

First express $\begin{pmatrix} 1 \\ 7 \\ 2 \\ 0 \end{pmatrix}$ as a linear combination $x \begin{pmatrix} 3 \\ 5 \\ 2 \\ 2 \end{pmatrix} + y \begin{pmatrix} 4 \\ 4 \\ 2 \\ 3 \end{pmatrix}$. We get $x = 3, y = -2$.

Then compute $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ as a linear combination $x \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Q10. Suppose A is a 4 x 4 matrix with exactly two distinct eigenvalues and three linearly independent eigenvectors. Which of the following are true?

- A. At least one of the eigenvalues has multiplicity equal to 2
- B. One of the eigenspaces has dimension 3.
- C. One of the eigenspaces has dimension 1.
- D. We can find a 4 x 3 matrix P that diagonalizes A.
- E. At most one of the eigenvalues has multiplicity equal to 1.

Explanation

Since there are only 3 linearly eigenvectors shared by the two eigenvalues, one eigenspace must have dimension 2 and the other eigenspace must have dimension 1. So option B is incorrect while option C is correct.

The multiplicities of the two eigenvalues can be either 2,2 or 3, 1. So option A is not always true and option E is always true.

Option D is incorrect since A is not diagonalizable. The matrix P that diagonalizes a matrix must be a square matrix.

Q11. Suppose A is a 3 x 3 diagonalizable matrix with two eigenspaces

$$E_1 = \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\} \text{ and } E_2 = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Which of the following matrices diagonalize A?

A. $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 2 & 1 & 0 \end{pmatrix}$

B. $\begin{pmatrix} 2 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

C. $\begin{pmatrix} 1 & 1/2 & 0 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}$

D. $\begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & -1 \end{pmatrix}$

E. $\begin{pmatrix} 1 & 1 & 0 \\ 3 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix}$

Explanation

A matrix that diagonalizes A must have two columns coming from E_1 and one column coming from E_2 . More specifically, two columns must be linear combinations of

$\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and one column must be scalar multiple of $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

Q12. An SDE has a solution given by

$$y = e^{2t} \left[\begin{pmatrix} -2 \\ 1 \end{pmatrix} \sin t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos t \right].$$

- A. The trajectories are spiralling clockwise towards the origin
- B. The trajectories are spiralling clockwise away from the origin
- C. The trajectories are spiralling anti-clockwise towards the origin
- D. The trajectories are spiralling anti-clockwise away from the origin**
- E. The trajectories are close loops moving clockwise around the origin

Explanation

From the given solution, an eigenvalue of the underlying matrix A is $r = 2 + i$ with corresponding eigenvector $v = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 + i \\ 1 \end{pmatrix}$.

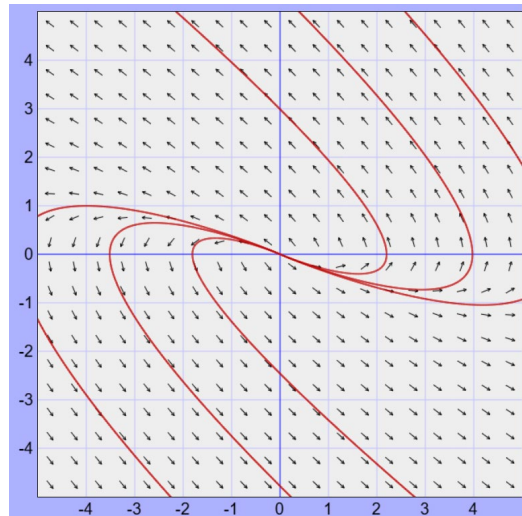
This gives an unstable spiral point, so the trajectories are moving away from the origin.

To determine the orientation of the trajectories, we find the matrix $A = PDP^{-1}$ where

$P = \begin{pmatrix} -2 + i & -2 - i \\ 1 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} 2 + i & 0 \\ 0 & 2 - i \end{pmatrix}$. This gives $A = \begin{pmatrix} 0 & -5 \\ 1 & 4 \end{pmatrix}$.

The test point $(1,0)$ has a tangent vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ pointing upwards. So the trajectories are moving anticlockwise.

Q13. The following is a phase portrait of a 2×2 SDE $y' = Ay$. Which of the following statements are true?



- A. The SDE has only one linearly independent solution.
- B. A has a repeated eigenvalue
- C. A has complex eigenvalues
- D. The two ends of every trajectory are parallel to the same vector
- E. There is no solution of the SDE that correspond to a straight-line trajectory

Explanation

The phase portrait shows that the equilibrium point is a degenerate node. This is the case when A has a repeated eigenvalue and only one linearly independent eigenvector. So option B is correct and option C is incorrect.

While there is only one linearly independent eigenvector for A , we can still form two linearly independent solutions for the SDE using the generalised eigenvector. So option A is incorrect.

Option D is correct: the two ends of every trajectory will be parallel to the guiding axis which is determined by the only eigenvector.

Option E is incorrect: There is a pair of straight line trajectories along the guiding axis.

Q14. Suppose $y' = Ay$ is a 2×2 SDE and A has two eigenvalues $r_1 = 0$ and $r_2 \neq 0$ with eigenvectors $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} c \\ d \end{pmatrix}$ respectively. Which of the following are solutions of the SDE?

- A. $y = \begin{pmatrix} ka \\ kb \end{pmatrix}$ for any k
- B. $y = \begin{pmatrix} a \\ b \end{pmatrix} + t \begin{pmatrix} c \\ d \end{pmatrix}$

$$C. y = \begin{pmatrix} ce^{r_2 t} \\ de^{r_2 t} \end{pmatrix}$$

$$D. y = e^{r_2 t} \begin{pmatrix} a + tc \\ b + td \end{pmatrix}$$

$$E. y = \begin{pmatrix} 2a \\ 2b \end{pmatrix} - e^{r_2 t} \begin{pmatrix} c \\ d \end{pmatrix}.$$

Explanation

From the given eigenvalues and eigenvectors, a fundamental set of solutions is given by

$Y_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $Y_2 = \begin{pmatrix} c \\ d \end{pmatrix} e^{r_2 t}$. Any linear combination of Y_1 and Y_2 will give a solution of the SDE.

Option A is $kY_1 + 0 Y_2$; Option C is $0Y_1 + 1 Y_2$; Option E is $2Y_1 - 1 Y_2$.

Option B and D are not given by such linear combinations.

Q15. FIB - Three tanks filled with salt water are connected by pipes, where water flows in and out among the tanks at some constant rates. Let y_1, y_2, y_3 be the amount of salt in tanks 1, 2 and 3 respectively at time t . The system is described by the 3x3 SDE:

$$\begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \frac{1}{100} \begin{pmatrix} -6 & 4 & 2 \\ 3 & -8 & 5 \\ 3 & 4 & -7 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.$$

Suppose the respective amount of salt in tank 1, 2 and 3 are 10kg, 20kg and 30kg initially, and a, b, c (in kg) in the long run. Then a, b, c are ... (Type the number only, no "kg")

Ans: a = 20, b = 20, c = 20

Explanation

The eigenvalues of the underlying matrix are given by 0, -3/25 and -9/100 with

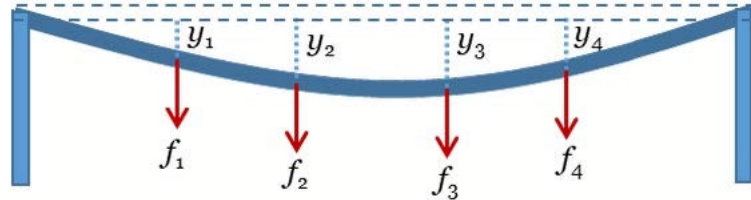
corresponding eigenvectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.

The general solution is given by $\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^{-3t/25} + c_3 \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} e^{-9t/100}$.

In the long run, the solution tends to $\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, which means that the amount of salt in the three tanks will be equal. So the initial total amount of salts $10+20+30 = 60\text{kg}$ will be evenly distributed to the three tanks.

Question 16 (10 marks)

A horizontal elastic beam is supported at each end and is subjected to downward forces f_1, f_2, f_3, f_4 (in newton) at four points as shown in the diagram, with amount of deflection y_1, y_2, y_3, y_4 (in cm) downward respectively.



The flexibility matrix is given by $D = \begin{pmatrix} 0.004 & 0.003 & 0.002 & 0.001 \\ 0.001 & 0.004 & 0.003 & 0.002 \\ 0.002 & 0.001 & 0.004 & 0.003 \\ 0.003 & 0.002 & 0.001 & 0.004 \end{pmatrix}$.

(i) (3 marks) If downward forces of 10 newton each are applied at the four points correspondingly, find the respective deflections at these points.

Answer and working

$$y = Df = \begin{pmatrix} 0.004 & 0.003 & 0.002 & 0.001 \\ 0.001 & 0.004 & 0.003 & 0.002 \\ 0.002 & 0.001 & 0.004 & 0.003 \\ 0.003 & 0.002 & 0.001 & 0.004 \end{pmatrix} \begin{pmatrix} 10 \\ 10 \\ 10 \\ 10 \end{pmatrix} = \begin{pmatrix} 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{pmatrix}$$

So $y_1 = y_2 = y_3 = y_4 = 0.1 \text{ cm}$.

(ii) (4 marks) Is it possible to apply forces to the four points such that there are no deflection at the first three points and a deflection of 0.2cm downward at the fourth point? If yes, find the respective forces; if no, justify your answer.

Answer and working

$$\text{Yes. } \mathbf{f} = \mathbf{D}^{-1}\mathbf{y} = \begin{pmatrix} 275 & -225 & 25 & 25 \\ 25 & 275 & -225 & 25 \\ 25 & 25 & 275 & -225 \\ -225 & 25 & 25 & 275 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ -45 \\ 55 \end{pmatrix}$$

$f_1 = f_2 = 5$ newton (downward); $f_3 = 45$ newton (upward); $f_4 = 55$ (downward)

(iii) (3 marks) Is it possible to apply two different combinations of forces f_1, f_2, f_3, f_4 at the 4 points to have the same combination of deflections y_1, y_2, y_3, y_4 at the respective points? Justify your answer.

Answer and working

No. The matrix D is non-singular. So the system $Df = y$ (for fixed y) must have a unique solution for f .

Question 17 (12 marks)

Let P_1 and P_2 be two planes in the XYZ-space with equations $x + y - 2z = 0$ and $2x - y + z = 0$ respectively.

(i) (3 marks) Form a linear system with the two equations and find the solutions of the system.

Answer and working

$$\begin{cases} x + y - 2z = 0 \\ 2x - y + z = 0 \end{cases}$$

Solve the system using Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 2 & -1 & 1 & 0 \end{array} \right) \xrightarrow{GE} \left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & -3 & 5 & 0 \end{array} \right)$$

General solution: $\begin{cases} x = t/3 \\ y = 5t/3 \\ z = t \end{cases}$

(ii) (2 marks) Express the line of intersection (L_1) in terms of a linear span. Explain how your answer is derived.

Answer and working

The general solution in (a) represents the line of intersection of the two planes.

Take any solution, say $x = 1$, $y = 5$, $z = 3$ (with $t = 3$). Then the line L_1 can be expressed as $\text{span}\{ (1, 5, 3) \}$ (can use any scalar multiple of $(1, 5, 3)$)

(iii) (3 marks) Let P_3 be a 3rd plane with equation $x + y + z = 0$. Express P_3 in terms of a linear span. Explain how your answer is derived.

Answer and working

We find any two vectors not parallel to each other that lie on the plane.

For example, $x = 1, y = -1, z = 0$ and $x = 0, y = 1, z = -1$ both satisfy the equation $x+y+z=0$. So the two vectors $(1,-1,0)$ and $(0,1,-1)$ lie on the plane and are not scalar multiple of each other. So the plane P_3 can be expressed as $\text{span}\{ (1,-1,0), (0, 1, -1) \}$.

(iv) (4 marks) Let L_2 be the line obtained by projecting L_1 onto P_3 . Find the line (L_2) in terms of a linear span. Show your working.

Answer and working

We find the projection of any vector in L_1 , say $v = (1, 5, 3)$, and project it onto the plane P_3 . To find the projection, we first find the least squares solution of $Ax = v$ where A is the 2×3 matrix with the two columns given by the spanning vectors in (c):

$$A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \text{ and } v = \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}.$$

$$\text{Solving } A^T A x = A^T v: \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}, \text{ we get } x_0 = \begin{pmatrix} -2 \\ 0 \end{pmatrix}.$$

$$\text{So the projection is } Ax_0 = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}$$

and the line L_2 can be expressed as $\text{span}\{(-2, 2, 0)\}$.

Question 18 (10 marks)

A certain species of animals has two life stages: juvenile (up to 1 year old) and adult. Each year,

- an adult produces an average of 2 juveniles;
- half of the juveniles survive to become adults;
- 5/6 of the adults survive.

(i) (3 marks) Find the stage matrix **A** such that $\begin{pmatrix} j_{n+1} \\ a_{n+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n .

Answer and working

$$\begin{aligned} j_{n+1} &= 0j_n + 2a_n \\ a_{n+1} &= \frac{1}{2}j_n + \frac{5}{6}a_n \end{aligned} \Rightarrow \begin{pmatrix} j_{n+1} \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1/2 & 5/6 \end{pmatrix} \begin{pmatrix} j_n \\ a_n \end{pmatrix}. \text{ So } A = \begin{pmatrix} 0 & 2 \\ 1/2 & 5/6 \end{pmatrix}.$$

(ii) (4 marks) Suppose the population of juveniles is 12000 in year 2025. What should the population of adults be in the same year so that the proportion of juveniles to adults remain the same in the following years? Explain how your answer is derived.

Answer and working

A has two eigenvalues $-2/3$ and $3/2$ with eigenvectors $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 4/3 \\ 1 \end{pmatrix}$ respectively.

In order for the proportion of the two populations to remain the same, j_{2025} and a_{2025} must form an eigenvector of A, which in turn has to be a scalar multiple of $\begin{pmatrix} 4/3 \\ 1 \end{pmatrix}$, since the other eigenvector is not applicable.

So given $j_{2025} = 12000$, we need $a_{2025} = \frac{3}{4} \times 12000 = 9000$.

(iii) (3 marks) Will the populations of the juveniles and adults stabilize in the long run? Justify your answer.

Answer and working

No. The dominating eigenvalue is $3/2$, which is greater than 1. In the long run the proportion of the two populations will stabilize in the ratio of 4:3, but both populations will increase by 1.5 times per year.

Question 19 (13 marks)

The oscillation of a damped mass-spring system is described by an ODE

$$my'' + cy' + ky = 0$$

where y is the (vertical) position of the ball, m is the mass of the ball in kg, k is the spring constant in Newton/meter, and c is the damping constant. Suppose the general solution of the ODE is given:

$$y = c_1 e^{-2t} + c_2 e^{-t/2}.$$

(i) (3 marks) Find the general solution of the associated SDE for this mass-spring system.

Answer and working

SDE is given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -k/m & -c/m \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$

We are given $y_1 = y = c_1 e^{-2t} + c_2 e^{-t/2}.$

By differentiating y w.r.t. t , we get $y_2 = y' = -2c_1 e^{-2t} - \frac{1}{2}c_2 e^{-t/2}.$

So the general solution of the SDE is given by

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 e^{-2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 e^{-t/2} \begin{pmatrix} 1 \\ -1/2 \end{pmatrix}$$

Alternatively, the general solution can also be presented as:

$$y_1 = c_1 e^{-2t} + c_2 e^{-t/2}$$

$$y_2 = -2c_1 e^{-2t} - \frac{c_2}{2} e^{-t/2}$$

(ii) (2 marks) Determine the type and stability of the equilibrium point of the SDE.

Answer and working

From the general solution, we have the two eigenvalues -2 and $-1/2$.

So the equilibrium point is a node and is asymptotically stable.

(iii) (2 marks) Suppose the ball is displaced from its rest position, describe in words the movement of the ball before it comes to an equilibrium state.

Answer and working

In this case, the system is over-damping (the damping force is strong relative to the spring force). The ball again will return to the equilibrium state slowly without oscillating.

(iv) (3 marks) Suppose $m = 2$. Find c and k .

Answer and working

From the general solution of the SDE in (i), the eigenvectors of -2 and $-1/2$ are

$\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1/2 \end{pmatrix}$ respectively.

We form the matrices $P = \begin{pmatrix} 1 & 1 \\ -2 & -1/2 \end{pmatrix}$ and $D = \begin{pmatrix} -2 & 0 \\ 0 & -1/2 \end{pmatrix}$

and find the underlying matrix

$$A = PDP^{-1} = \begin{pmatrix} 1 & 1 \\ -2 & -1/2 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & -1/2 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & -2.5 \end{pmatrix}.$$

So $-k/m = -1 \rightarrow k = 2$ and $-c/m = -2.5 \rightarrow c = 5$.

(v) (3 marks) Sketch the phase portrait of the SDE.

Answer and working

